TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Investigation of the Phenomenon of Polarization

Polarizer and analyser, crossed extinction, Malus's law verified point by point, and polarization by reflection at Brewster's angle — the experiments that prove light is a transverse wave.

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 17 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to investigate polarization qualitatively and quantitatively. Ordinary light vibrates in all directions perpendicular to its travel. A **polarizer** (Polaroid sheet or a tourmaline/Nicol device) transmits only the component vibrating along its transmission axis, producing **plane-polarized** light. A second sheet, the **analyser**, tests it: as the analyser rotates through angle θ to the polarizer's axis, the transmitted intensity follows **Malus's law**

$$I = I_0 \cos^2\theta$$

falling to zero at θ = 90° — **crossed polarizers extinguish the beam**, which no longitudinal wave could do. Light reflected from a dielectric surface is also partially polarized, and at **Brewster's angle** θ_B it is completely plane-polarized, with tanθ_B = μ (Physics-I Topic 6).

2. Apparatus and procedure

- **Apparatus:** two Polaroid sheets in rotating mounts with angle scales, a light source and a light meter (or a photocell/galvanometer), a glass plate on a rotating table for the reflection investigation.
- **Crossing test:** view the source through one sheet (note the dimming, about half). Add the second sheet and rotate it: record the brightness cycle — maximum, extinction at 90°, maximum again at 180°. This alone demonstrates polarization.
- **Malus's law:** fix the polarizer; rotate the analyser in 15° steps; read the meter at each θ; tabulate I against cos²θ; plot — a straight line through the origin verifies the law.
- **Reflection:** reflect a beam from the glass plate onto the analyser; rotate the plate's incidence angle and the analyser together: near θ_B (56° for glass) the reflected beam is extinguished at one analyser orientation — completely polarized — while away from θ_B extinction is only partial.

- **Surprises to try:** a third sheet inserted between crossed polarizers at 45° lets light through again; cellophane or a stressed transparent plastic between crossed polarizers shows colours (birefringence) — record both observations.

3. Worked verification (model data)

MALUS'S LAW CHECK

Θ	$\cos^2\Theta$	I MEASURED (RELATIVE)	I PREDICTED = $I_0\cos^2\Theta$
0°	1.000	1.00	1.00
30°	0.750	0.74	0.75
45°	0.500	0.51	0.50
60°	0.250	0.26	0.25
90°	0.000	0.01 (leakage)	0.00

Measured intensities track $\cos^2\theta$ within the meter's precision — the law is verified, and the tiny residue at 90° honestly records the Polaroid's imperfection. Brewster check on the same bench: extinction of the reflected beam at incidence 56.3° gives $\mu = \tan 56.3° = 1.50$, the glass's index (compare Experiment 13's method).

4. Precautions and practical notes

- Exclude ambient light: the meter reads room light too, and at the analyser's dark settings stray light dominates the residue.
- Rotate the analyser in one direction and read at settled positions; keep the source intensity steady (lamp warmed up).
- The angle θ is between the two sheets' transmission axes — find each sheet's axis first from the extinction setting, not from the mount's printed zero.
- For the reflection part, keep the reflected beam centred on the analyser as the plate rotates (the beam walks as the angle changes).
- Record imperfections (leakage at crossing, incomplete extinction off-Brewster) as observations, not embarrassments — they measure the polarizer's and the surface's quality.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why does the extinction of light by crossed polarizers prove that light is a transverse wave?
2. Unpolarized light of intensity I passes a polarizer, then an analyser at 30° to it. What fraction emerges?
3. State Malus's law and explain, from the law, why the transmitted intensity through a single polarizer is half the incident unpolarized intensity.
4. How can crossed polarizers with a third sheet at 45° between them pass light, when the outer two are crossed?
5. Glass of $\mu = 1.60$ is used in the reflection investigation. At what incidence is the reflected beam completely polarized?
6. Give two everyday applications of polarization met in this investigation's phenomena.

Answer key

1. Because a longitudinal wave (vibration along the direction of travel) has no sideways orientation for a sheet to select or block — rotating an analyser could not extinguish it. Only a transverse wave, vibrating in a plane the polarizer can align with or against, can be extinguished by rotation: polarization is the signature of transverseness.
2. The polarizer passes half: I/2 (plane-polarized). The analyser then applies Malus: $(I/2)\cos^2 30° = (I/2)(0.75) = 0.375\ I$ — three-eighths of the original intensity emerges.
3. $I = I_0 \cos^2\theta$, where I_0 is the intensity of the plane-polarized beam incident on the analyser. For unpolarized light through the first polarizer, the transmitted component is the average of $\cos^2$ over all vibration directions, and $\langle \cos^2 \rangle = ½$ — hence exactly half passes, now plane-polarized.
4. The middle sheet re-orients the light: it transmits the component of the (first sheet's) polarized light along its own 45° axis — reducing but not extinguishing it — and that transmitted light is now polarized at 45°, so it has a component along the last sheet's axis and partly passes. Each sheet only ever removes the component across its own axis; the middle one manufactures a component the final sheet can pass.
5. $\theta_B = \tan^{-1}\mu = \tan^{-1}(1.60) = 58.0°$.
6. Any two of: Polaroid sunglasses and camera filters cutting reflected glare (they block the horizontally polarized reflection from roads and water); LCD screens, which switch pixels by rotating polarized light; stress analysis of transparent models (photoelastic colours between crossed polarizers reveal strain); 3D cinema glasses separating the two eye-images by polarization.